

Build & Evaluate a Linear Regression Model (House Price Predictor)

Artificial Intelligence & Machine Learning – Task 1

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Date: 22/7/2026

1. Objective

The objective of this project is to develop and evaluate a Linear Regression model using the California Housing dataset. This project demonstrates the complete machine learning workflow including data loading, preprocessing, exploratory data analysis (EDA), model development, prediction, evaluation and reporting.

2. Introduction

Machine Learning is a branch of Artificial Intelligence that enables systems to learn from historical data. Linear Regression is one of the most widely used supervised learning algorithms for predicting continuous values. In this project, the California Housing dataset is used to predict median house prices using eight numerical input features.

3. Dataset Description

The California Housing dataset is available through Scikit-learn and contains 20,640 records with 9 columns including the target variable MedHouseVal. The dataset contains attributes such as Median Income, House Age, Average Rooms, Average Bedrooms, Population, Average Occupancy, Latitude and Longitude.

4. Libraries Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Jupyter Notebook

5. Exploratory Data Analysis

EDA was performed using `df.head()`, `df.info()` and `df.describe()` to understand the dataset. Correlation analysis was performed using a heatmap. No significant missing values were observed. The distributions of numerical features were analysed before model training.

Insert Screenshot: `df.head()`

Insert Screenshot: `df.info()`

Insert Screenshot: `df.describe()`

Insert Screenshot: Correlation Heatmap

6. Model Building

The dataset was divided into training and testing sets using an 80:20 split. The Linear Regression algorithm from Scikit-learn was trained using the training dataset. Predictions were generated for unseen testing data and evaluated using MAE, RMSE and R^2 Score.

7. Model Evaluation

The model achieved the following evaluation metrics:

MAE = 0.5332

RMSE = 0.7456

R^2 Score = 0.5758

These values indicate reasonable prediction performance for a baseline Linear Regression model.

Insert Screenshot: Actual vs Predicted Plot

Insert Screenshot: Residual Plot

8. Discussion

The predicted values closely followed the actual values, although some prediction error remained because Linear Regression assumes a linear relationship among variables. The model successfully demonstrates the end-to-end regression workflow.

9. Conclusion

The project successfully implemented a Linear Regression model for predicting California housing prices. It covered dataset loading, EDA, preprocessing, model training, prediction and evaluation. The project improved practical understanding of supervised machine learning.

10. Future Improvements

- Random Forest Regression
- XGBoost
- Feature Engineering
- Hyperparameter Tuning
- Outlier Handling
- Cross Validation
- Deploy using Flask or Streamlit

11. References

- Scikit-learn Documentation
- California Housing Dataset
- Python Documentation
- Pandas Documentation